

MS4215: Advanced Data Analysis

Week 2A: Correlation and Simple Linear Regression

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Learning outcomes

- Calculate and interpret the Pearson correlation coefficient, including its sign, magnitude, and limitations
- Fit a simple linear regression model in R using `lm()` and interpret the intercept and slope
- Decompose total variation using $SST = SSR + SSE$ and interpret R^2 as the proportion of variance explained
- State the LINE assumptions for OLS regression and evaluate them using R diagnostic plots

Motivating question

From Week 1B, in the Irish weather data, we could see that `tmax_c` and `tmin_c` tend to move together.

We will also examine a new dataset today asking whether advertising spend predicts sales.

Question 1

How do we *measure* how strongly two variables are associated?

→ **Correlation** (Section 1)

Question 2

Can we build an *equation* to predict one from the other?

→ **Regression** (Section 2)

How variables move together

Reading a scatter plot

When two quantitative variables are plotted together, look for three things:

Direction

Positive: as $X \uparrow$, $Y \uparrow$

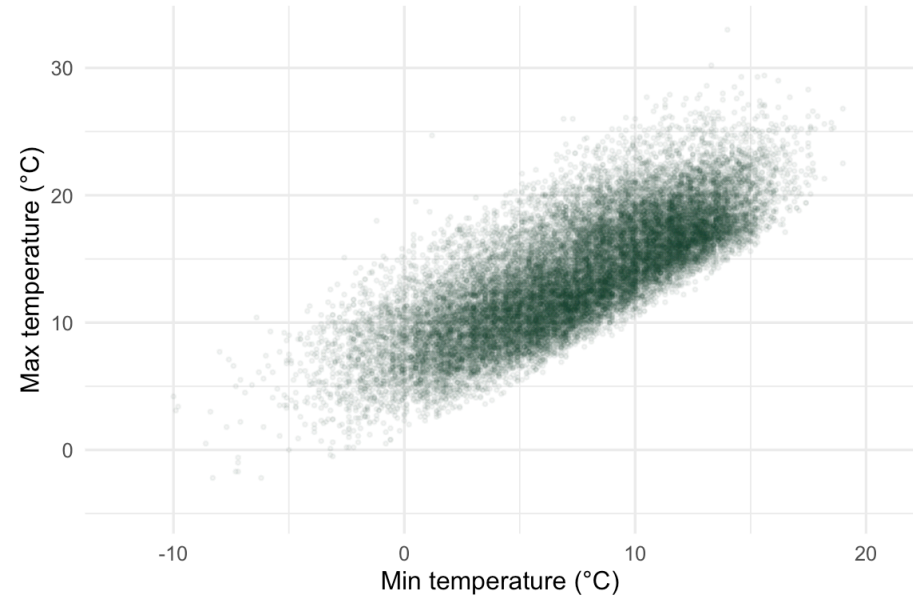
Negative: as $X \uparrow$, $Y \downarrow$

Form

Linear or non-linear (curved)?

Strength

How tightly do the data cluster around the pattern?



Note

Always plot before you calculate. If the scatter shows a curve, Pearson r will mislead you.

The Pearson correlation coefficient

The **population** correlation between X and Y:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The **sample** estimate, given data $x_1, \dots, x_n$ and $y_1, \dots, y_n$:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \times \sum_{i=1}^n (y_i - \bar{y})^2}}$$

r is the sample estimate of the unknown population parameter ρ .

Properties of r

- $r \in [-1, 1]$ always
- **Sign** indicates direction: positive $\rightarrow$ variables rise together; negative $\rightarrow$ one rises as the other falls
- **Magnitude** indicates strength: $|r|$ closer to 1 = stronger linear association
- $r = \pm 1$ only for a **perfect** linear relationship (all points exactly on a line)
- $r = 0$ means **no linear** association: there may still be a strong curved relationship
- r is **dimensionless**: changing units of X or Y does not change r

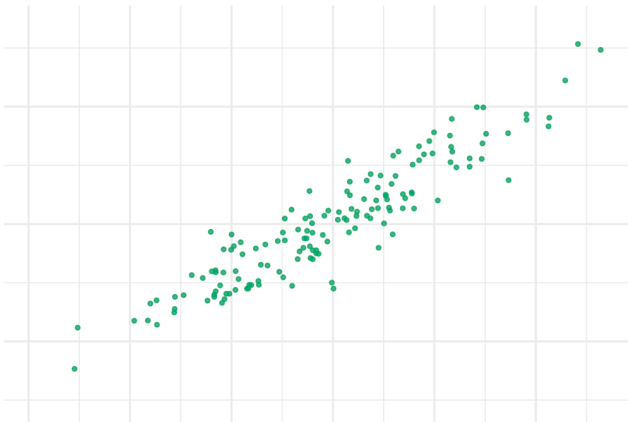
$ r $	Interpretation
0.00	None
0.01 - 0.19	Very weak
0.20 - 0.39	Weak
0.40 - 0.69	Modest
0.70 - 0.89	Strong
0.90 - 0.99	Very strong
1.00	Perfect

Warning

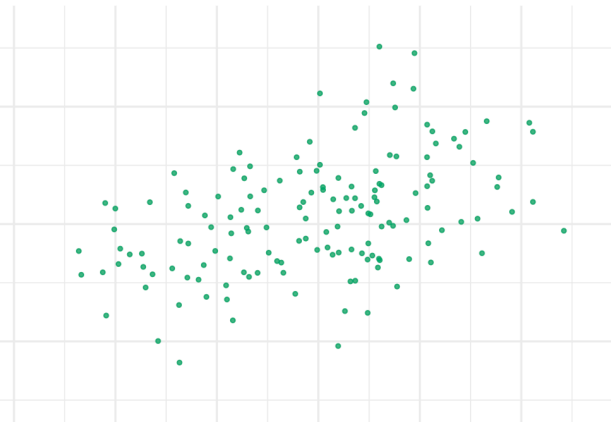
$r = 0$ means **no linear** association. Not no association at all.

Gallery of r values

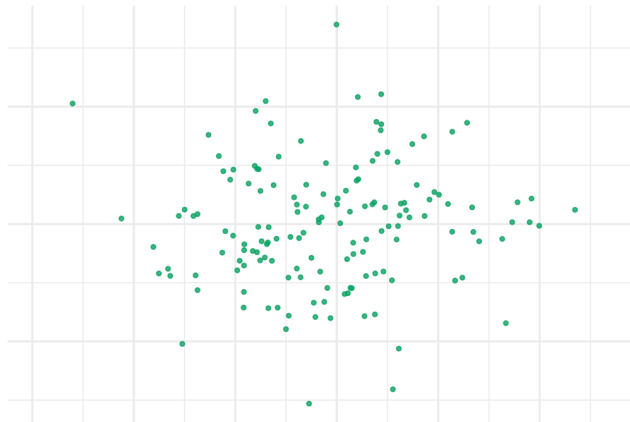
$r = 0.9$



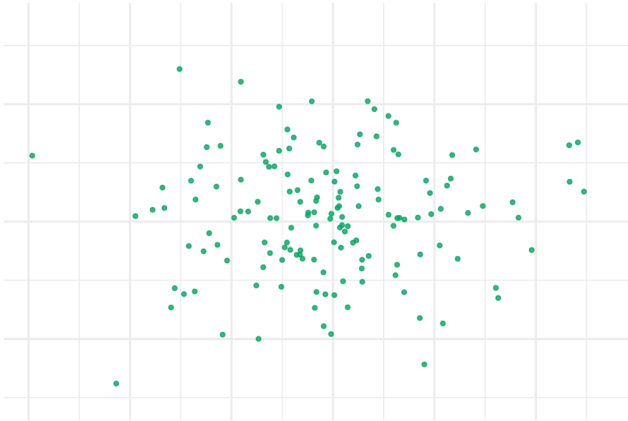
$r = 0.5$



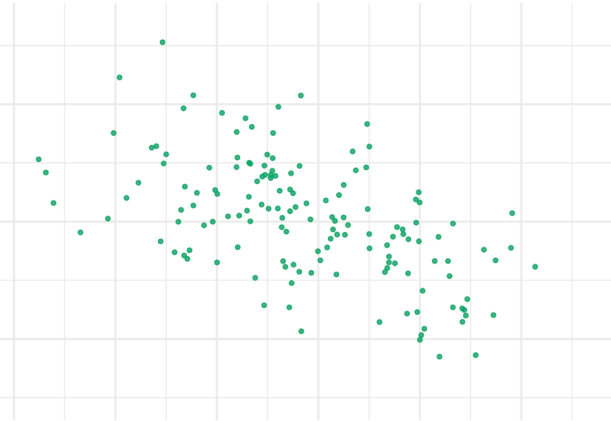
$r = 0.1$



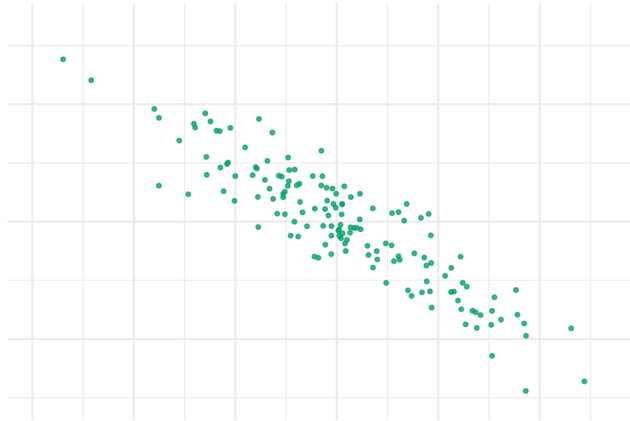
$r = -0.1$



$r = -0.5$



$r = -0.9$



Irish weather: Computing r

```
1 # tmax_c and tmin_c
2 cor(ireland$tmax_c, ireland$tmin_c, use = "complete.obs")
[1] 0.7809541
```

```
1 # tmax_c and rainfall
2 cor(ireland$tmax_c, ireland$prcp_mm, use = "complete.obs")
[1] -0.03002593
```

`use = "complete.obs"` drops rows with any `NA` before computing.

Without it, a single missing value returns `NA`.

Interpretation:

- $tmax_c \sim tmin_c \approx 0.78 \rightarrow$ **strong positive**
- $tmax_c \sim prcp_mm \approx -0.03 \rightarrow$ **negligible** (essentially no linear association)

Warmer days tend to have higher minimum temperatures.

Rainfall has almost no linear relationship with maximum temperature in this dataset.

Correlation matrix

```
1 ireland_sub <- ireland[, c("tmax_c", "tmin_c", "prcp_mm")]  
2  
3 cor(ireland_sub, use = "complete.obs")
```

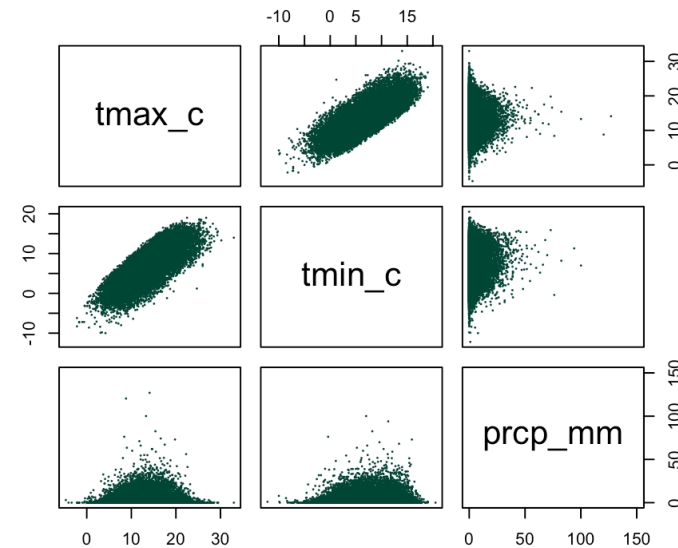
```
          tmax_c    tmin_c    prcp_mm  
tmax_c    1.00000000 0.78586057 -0.04539023  
tmin_c    0.78586057 1.00000000  0.05293042  
prcp_mm  -0.04539023 0.05293042  1.00000000
```

How to read it:

- **Diagonal:** always 1 (a variable is perfectly correlated with itself)
- **Off-diagonal:** pairwise Pearson r
- **Symmetric:** upper triangle = lower triangle ($r(X,Y) = r(Y,X)$)

pairs(): Scatter matrix

```
1 pairs(ireland[, c("tmax_c", "tmin_c", "prcp_mm")],  
2       col = "#034638",  
3       pch = 16,  
4       cex = 0.3)
```



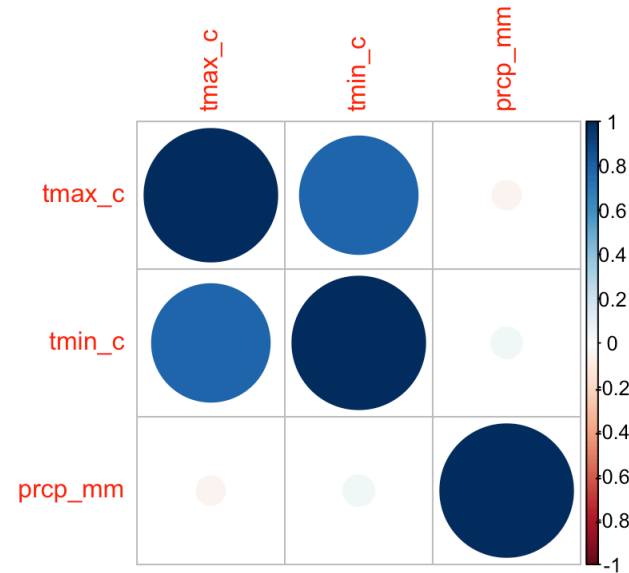
Nine mini scatter plots: every variable against every other.

The matrix and the visual tell the same story: read them together.

Upper and lower triangles are mirror images.

corrplot: Visual correlation matrix

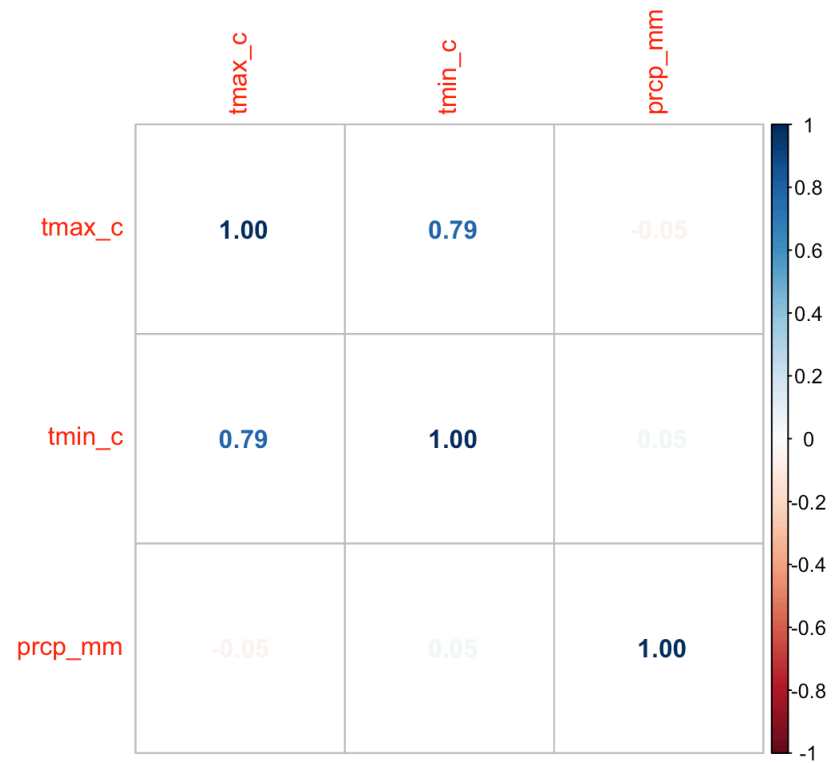
```
1 library(corrplot)
2
3 R_mat <- cor(
4   ireland[, c("tmax_c", "tmin_c", "prcp_mm")],
5   use = "complete.obs"
6 )
7 corrplot(R_mat)
```



Reading the plot:

- Circle **size**: larger = stronger $|r|$
- Circle **colour**: blue = positive, red = negative

```
1 # Show r values instead of circles
2 corrpplot(R_mat, method = "number")
```



Spearman rank correlation

```
1 # Pearson (default)
2 cor(ireland$tmax_c,
3     ireland$tmin_c,
4     use = "complete.obs")
```

```
[1] 0.7809541
```

```
1 # Spearman
2 cor(ireland$tmax_c,
3     ireland$tmin_c,
4     use = "complete.obs",
5     method = "spearman")
```

```
[1] 0.7912497
```

How Spearman works:

1. Replace each value with its **rank** (1 = smallest)
2. Compute Pearson r on the **ranks**

When to use Spearman:

- Ordinal data (e.g. Likert scales, degree class)
- Continuous data with **severe outliers**
- Heavy-tailed distributions
- Monotone but possibly non-linear relationships

Note

For Irish weather: Pearson $\approx$ Spearman, confirms the relationship is approximately linear with no extreme outliers dominating.

Pearson vs Spearman: When to choose

Use Pearson when:

- Both variables are continuous
- Distribution is approximately symmetric
- No severe outliers
- You expect a linear relationship

Default for continuous, well-behaved data.

Use Spearman when:

- One or both variables are ordinal
- Severe outliers are present
- Distribution is heavily skewed
- Relationship may be monotone but not linear

One extra argument: `method = "spearman"`

Note

When in doubt with continuous data, check both. If they agree, either is fine. If they diverge noticeably, investigate the reason.

Correlation $\neq$ causation

Three ways two variables can be correlated:

(a) Causation

X directly causes Y

$$X \longrightarrow Y$$

e.g. *smoking* $\rightarrow$ *lung cancer*

(b) Common response

A third variable Z drives **both** X and Y

$$Z \longrightarrow X, \quad Z \longrightarrow Y$$

$r(\text{ice cream, polio}) \approx +1$

Ice cream causes polio?

The real explanation:

Summer heat $\rightarrow$ ice cream sales $\uparrow$

Summer heat $\rightarrow$ children gathering
 $\rightarrow$ polio spread $\uparrow$

Ice cream **does not cause** polio.

(c) Pure coincidence

Data happen to co-move over the observed period. No shared cause, no mechanism.

Recall from Week 1B (source: tylervigen.com):

- Psychology graduates $\sim$ groundskeeping jobs in Utah
- Neptune's distance from Sun $\sim$ burglaries
- Searches for "Friends" $\sim$ Lululemon stock

Warning

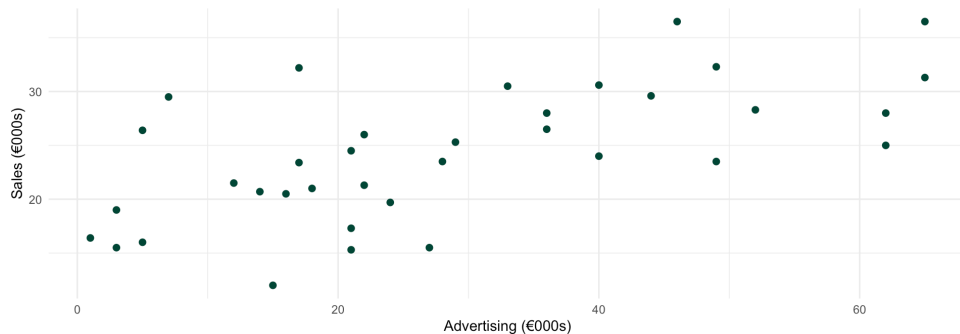
A correlation - no matter how strong - cannot tell you **why** two variables are associated.

Introducing the Sales dataset

```
1 sales <- read_csv("data/sales.csv")
2 head(sales, 4)
```

```
# A tibble: 4 × 2
  sales advert
<dbl> <dbl>
1 12      15
2 20.5    16
3 21      18
4 15.5    27
```

```
1 ggplot(sales, aes(x = advert, y = sales)) +
2   geom_point(colour = "#034638", size = 2) +
3   labs(x = "Advertising (€000s)",
4        y = "Sales (€000s)") +
5   theme_minimal()
```



Dataset: a sample of $n = 36$ advertising campaigns.

- **advert** : money spent on advertising (€000s)
- **sales** : resulting sales revenue (€000s)

From the scatter:

- Direction: **positive**
- Form: broadly **linear**
- Strength: **moderate**, clear trend with noticeable scatter

This scatter shows that more advertising is associated with higher sales, but the relationship is not perfect.

cor() for Sales

```
1 cor(sales$advert, sales$sales)
[1] 0.6314276
```

$$r = 0.631$$

From the interpretation table: **Modest** (0.40 - 0.69).

What r tells us:

- The association is **positive**
- It is **moderate** in strength, real but not near-perfect

What r does not tell us:

- How much sales increase per extra €1,000 of advertising
- Whether the relationship could be (maybe) causal
- The equation for prediction

Note

For prediction, we need regression. Not just a correlation coefficient.

What correlation cannot tell us

A strong r tells us that two variables are **associated linearly**. It does not tell us:

It cannot tell us:

- Which variable drives the other
- How much Y changes per unit increase in X
- The equation to predict Y from X
- Whether the relationship holds outside the observed range
- Whether the relationship is causal

Example:

$r = 0.631$ for Sales $\sim$ Advert.

We know the association is positive and moderate.

But the manager asks: *“By how much do sales increase per extra €1,000 of advertising?”*

r cannot answer this.

→ **Regression** equation can

Correlation: toolkit summary

Goal	Tool
Explore association between two continuous variables	Scatter plot: <code>ggplot</code> or <code>plot()</code>
Measure direction and strength (linear, no outliers)	<code>cor(x, y)</code> , Pearson
Robust to outliers or ordinal data	<code>cor(x, y, method = "spearman")</code>
Explore many pairs at once	<code>pairs()</code> or <code>corrplot()</code>
Handle missing values	add <code>use = "complete.obs"</code>

Always plot first.

Always interpret cautiously.

Correlation $\neq$ causation.

From association to prediction

The regression question

$r = 0.631$ for Sales $\sim$ Advert.

We know the association is **positive** and **moderate**.

But the manager asks:

“If we spend €20,000 on advertising, what sales revenue can we expect?”

Regression answers this by fitting the **equation of a straight line** through the data:

$$\hat{y} = b_0 + b_1 x$$

The population model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

Each component:

- Y_i : response (sales revenue, campaign i)
- x_i : predictor (advertising spend, campaign i)
- β_0 : true intercept (unknown parameter)
- β_1 : true slope (unknown parameter)
- ε_i : error: everything **not** explained by x_i

An alternative way to write this:

$$Y_i = \underbrace{(\beta_0 + \beta_1 x_i)}_{\text{Signal (Model)}} + \underbrace{\varepsilon_i}_{\text{Noise}}$$

Assumptions on the errors:

- Mean zero: $E(\varepsilon_i) = 0$
- Constant variance: $\text{Var}(\varepsilon_i) = \sigma^2$
- Normally distributed
- Independent of each other

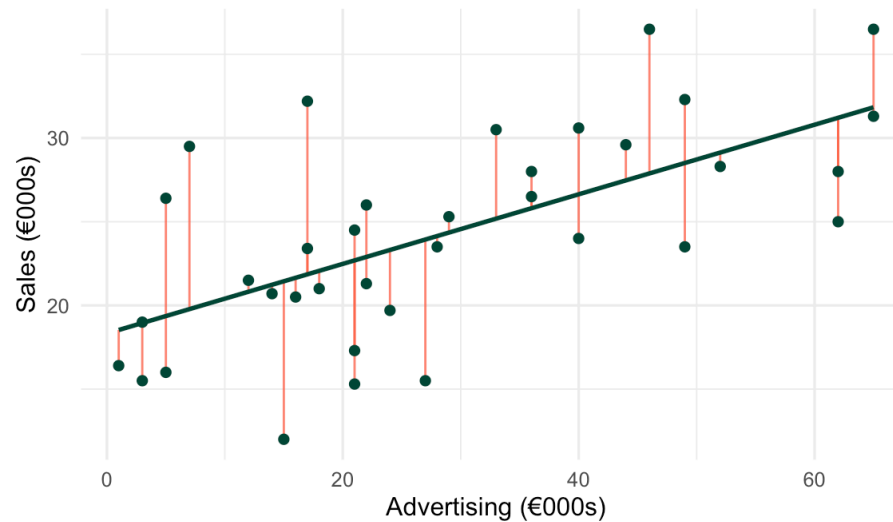
What Ordinary Least Squares (OLS) does

We cannot know the true β_0 and β_1 .

We **estimate** them: choose b_0 and b_1 that minimise the **sum of squared residuals**:

$$\min_{b_0, b_1} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

Ordinary Least Squares minimises the sum of squared vertical distances from each point to the fitted line, shown in red.



OLS formulae

The unique OLS estimates are:

$$\hat{b}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

where

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Two key properties of $\hat{b}_1$:

- $E(\hat{b}_1) = \beta_1$: the OLS slope is **unbiased**
- $\text{Var}(\hat{b}_1) = \frac{\sigma^2}{S_{xx}}$: SE is smaller when X values are more spread out

Note

The fitted line **always passes through** $(\bar{x}, \bar{y})$.

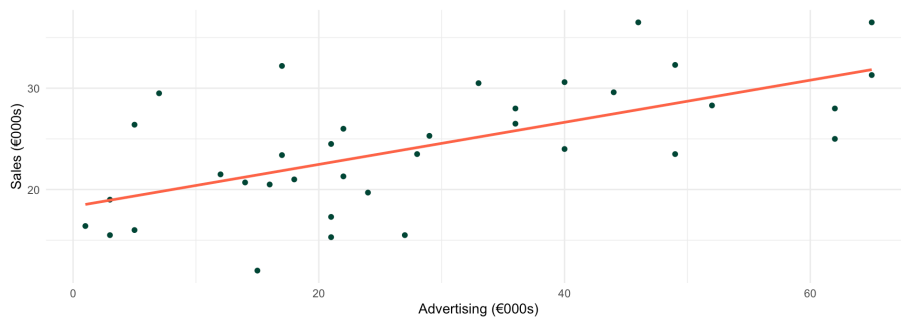
lm() in R

```
1 reg1 <- lm(sales ~ advert, data = sales)
```

R formula syntax: **response ~ predictor**

Add the fitted line to a scatter plot:

```
1 ggplot(sales,  
2       aes(x = advert, y = sales)) +  
3   geom_point(colour = "#034638") +  
4   geom_smooth(method = "lm",  
5             se      = FALSE,  
6             colour = "tomato") +  
7   labs(x = "Advertising (€000s)",  
8       y = "Sales (€000s)") +  
9   theme_minimal()
```



The **lm** object stores everything:

```
1 coef(reg1)      # b0 and b1  
2 fitted(reg1)    #  $\hat{y}_i$  values  
3 residuals(reg1) # ei values  
4 summary(reg1)   # full output  
5 anova(reg1)     # ANOVA table  
6 confint(reg1)   # CIs for  $\beta$ 
```

geom_smooth(method = "lm") is a visual shortcut.

Inference comes from **summary(reg1)**.

Fitted values and residuals

```
1 # Fitted values:  $\hat{y}_i = b_0 + b_1 \times x_i$ 
2 # Residuals:  $e_i = y_i - \hat{y}_i$ 
3
4 # First few rows: observed, fitted, residual
5 head(data.frame(
6   advert = sales$advert,
7   sales   = sales$sales,
8   fitted  = round(fitted(reg1), 2),
9   resid   = round(residuals(reg1), 2)
10 ))
```

	advert	sales	fitted	resid
1	15	12.0	21.44	-9.44
2	16	20.5	21.65	-1.15
3	18	21.0	22.06	-1.06
4	27	15.5	23.94	-8.44
5	21	15.3	22.69	-7.39
6	49	23.5	28.51	-5.01

$$\hat{y}_i = b_0 + b_1 x_i \quad (\text{on the line})$$

$$e_i = y_i - \hat{y}_i \quad (\text{vertical distance})$$

OLS minimises $\sum e_i^2$.

The **population error** ε_i is unknown and theoretical.

The **residual** e_i is observed and computed.

We use residuals to check whether the ε assumptions are plausible (Section 3).

A large $|e_i|$ = a point the model predicts poorly.

Interpreting the coefficients

From `summary(reg1)` :

$$\widehat{\text{Sales}} = 18.32 + 0.21 \times \text{advert}$$

Slope $b_1 = 0.21$:

For every extra €1,000 spent on advertising, predicted sales **increase by €210** (€0.21k) on average.

Intercept $b_0 = 18.32$:

Predicted sales when advertising spend = €0.

(May not be meaningful if zero lies outside the observed range of advert values.)

Warning

Interpretation template:

“For a one-unit increase in X, the predicted value of Y changes by b_1 units, *on average*.”

Always state the units.

- Units of b_1 : €000s sales per €000s advert
- Units of b_0 : €000s sales (when advert = 0)
- “On average” is essential. The slope is the change in the **conditional mean** of Y

summary() output

```
Call:
lm(formula = sales ~ advert, data = sales)

Residuals:
    Min       1Q   Median       3Q      Max
-9.4409 -3.2483 -0.5335  2.4216 10.3434

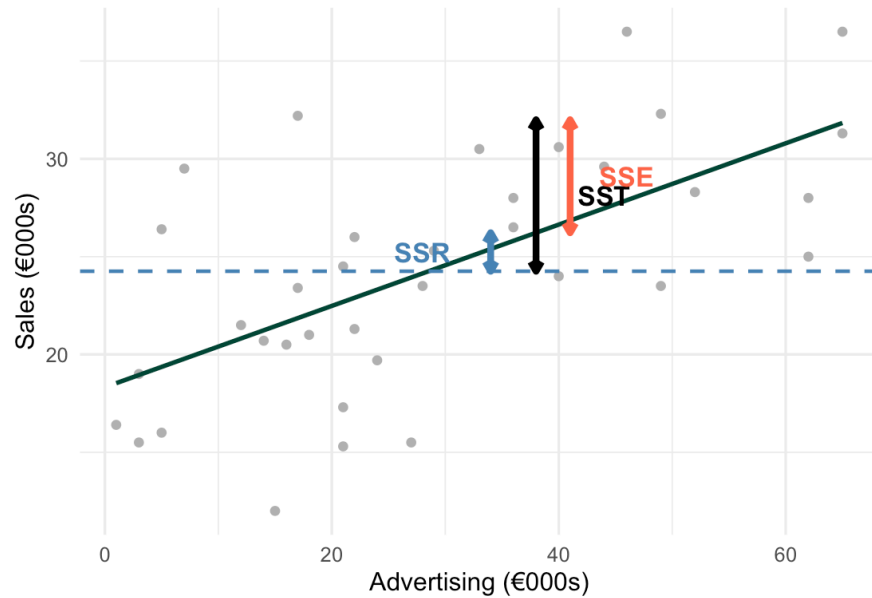
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 18.32299    1.48886   12.307 4.46e-14 ***
advert       0.20786    0.04378    4.748 3.64e-05 ***

Residual standard error: 4.863 on 34 degrees of freedom
Multiple R-squared:  0.3987,    Adjusted R-squared:  0.381
F-statistic: 22.54 on 1 and 34 DF,  p-value: 3.635e-05
```

Five things to locate every time:

- (1) Coefficients: Estimate, SE, t, p
- (2) Residual standard error $\hat{\sigma}$
- (3) R^2
- (4) Adjusted R^2
- (5) F-statistic

Variance decomposition: $SST = SSR + SSE$



$$SST = SSR + SSE$$

- **SST** (total sum of squares): total variation in Y from $\bar{y}$ (dashed line)
- **SSR** (regression sum of squares): variation *explained* by the regression line (blue)
- **SSE** (error sum of squares): unexplained variation, residuals (red)

R^2 : Coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

$R^2 \in [0, 1]$: the proportion of variance in Y explained by the regression.

For Sales ~ Advert:

```
1 summary(reg1)$r.squared
```

```
[1] 0.3987008
```

$R^2 = 0.399$: advertising spend explains **39.9%** of the variance in sales revenue.

The remaining 60.1% is due to other factors not captured in the model.

Note

In simple linear regression: $R^2 = r^2$. Check: $0.631^2 = 0.398 \approx 0.399 \checkmark$

The ANOVA table

```
1 anova(reg1)
```

Analysis of Variance Table

Response: sales

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
advert	1	533.20	533.20	22.544	3.635e-05 ***
Residuals	34	804.15	23.65		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Source	df	SS	MS	F
Regression (advert)	1	SSR	SSR/1	MSR/MSE
Residuals	$n - 2 = 34$	SSE	SSE/34	
Total	$n - 1 = 35$	SST		

F-test: $H_0 : \beta_1 = 0$ (advertising has no linear effect on sales)

$$F = \frac{\text{MSR}}{\text{MSE}} \sim F_{1, n-2} \text{ under } H_0$$

Large $F \rightarrow$ small $p \rightarrow$ reject H_0 .

For Sales: $F = 22.544$, $p = 3.635 \times 10^{-5}$.

Inference on β_1

t-test for the slope:

$$T = \frac{\hat{b}_1}{\text{SE}(\hat{b}_1)} \sim t_{n-2} \quad \text{under } H_0 : \beta_1 = 0$$

From `summary(reg1)` : $t = 4.748$, $p = 3.64 \times 10^{-5}$.

Strong evidence that $\beta_1 \neq 0$: advertising spend is a statistically significant predictor of sales.

95% confidence interval for β_1 :

```
1 confint(reg1)
```

	2.5 %	97.5 %
(Intercept)	15.2972692	21.3487059
advert	0.1188931	0.2968274

We are 95% confident the true slope lies between 0.12 and 0.30: for every extra €1,000 of advertising, sales increase by somewhere between €120 and €300 on average.

Prediction

```
1 # Point prediction: advert = 30 (€30,000 spend)
2 predict(reg1,
3         newdata = data.frame(advert = 30))
```

```
      1
24.55879
```

```
1 # 95% prediction interval
2 predict(reg1,
3         newdata = data.frame(advert = 30),
4         interval = "prediction")
```

```
      fit      lwr      upr
1 24.55879 14.53826 34.57933
```

Point prediction:

$$\hat{y} = 18.32 + 0.21 \times 30 = 24.55$$

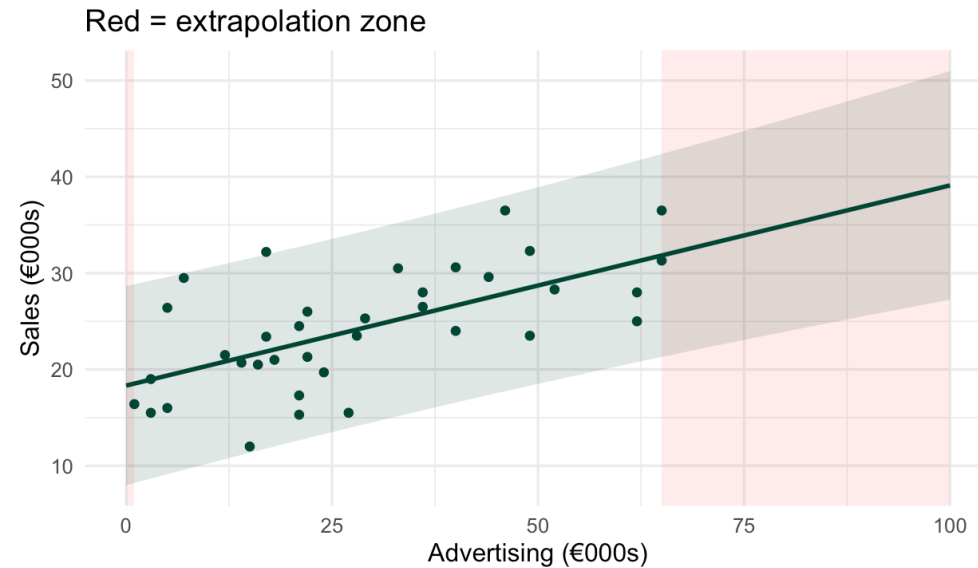
Predicted sales $\approx$ **€24,558**

95% prediction interval:

Captures where a **single new observation** at advert = 30 is expected to fall.

Always wider than a confidence interval for the mean response. It must account for both uncertainty in the line AND natural variability of individual observations (the σ^2 term).

Interpolation vs extrapolation



Interpolation: predict within the observed range of X = reliable.

Extrapolation: predict outside the range (red zones) = the linear relationship may not hold.

⚠ Warning

Never extrapolate without caution. The model was built on data within the observed range.

Applying to Irish weather

```
1 reg2 <- lm(tmax_c ~ tmin_c, data = ireland)
2 summary(reg2)
```

```
Call:
lm(formula = tmax_c ~ tmin_c, data = ireland)

Residuals:
    Min       1Q   Median       3Q      Max
-6.9617 -2.0542 -0.3518  1.7741 16.0028

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.72448    0.03547   217.8  <2e-16 ***
tmin_c       0.81062    0.00410   197.7  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.787 on 25006 degrees of freedom
(17182 observations deleted due to missingness)
Multiple R-squared:  0.6099,    Adjusted R-squared:  0.6099
```

From `summary(reg2)`:

Slope b_1 : for every 1°C increase in minimum temperature, predicted maximum temperature increases by approximately **0.81°C**

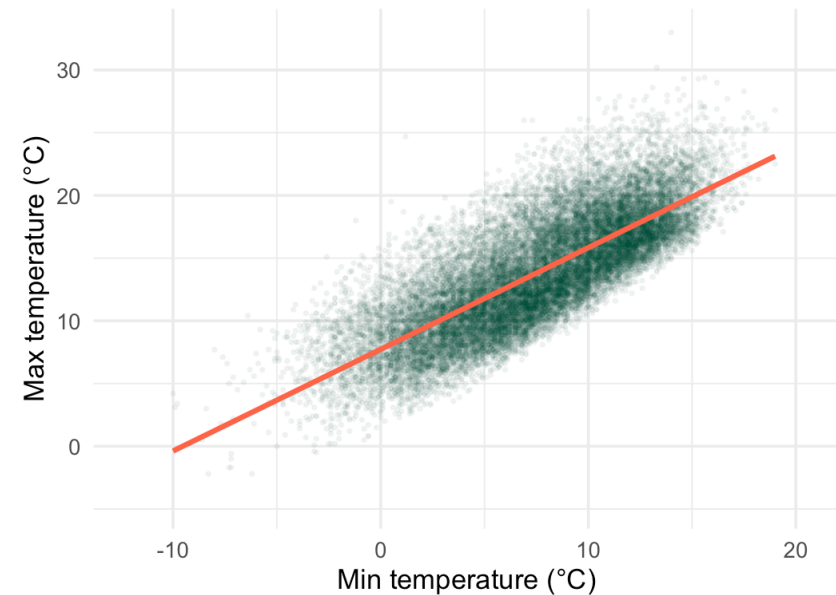
R²: minimum temperature explains **61%** of the variance in maximum temperature

Same workflow, new context.

- **Response Y:** `tmax_c` (daily maximum temperature, °C)
- **Predictor X:** `tmin_c` (daily minimum temperature, °C)
- $n = 25,008$ complete daily records from 5 Irish weather stations, 2000–2023

Irish weather: Scatter with fitted line

```
1 ggplot(ireland,  
2       aes(x = tmin_c, y = tmax_c)) +  
3   geom_point(alpha = 0.05,  
4             colour = "#034638",  
5             size = 0.5) +  
6   geom_smooth(method = "lm",  
7             se = TRUE,  
8             colour = "tomato",  
9             fill = "tomato",  
10            alpha = 0.25) +  
11   labs(x = "Min temperature (°C)",  
12        y = "Max temperature (°C)") +  
13   theme_minimal(base_size = 13)
```



With $n = 25,008$, the confidence band for the line is extremely narrow.

High R^2 expected, minimum temperature is a strong predictor of maximum temperature.

Interpreting reg2

From `summary(reg2)`, answer the same four questions:

Question	Where to look
Is the slope statistically significant?	$\Pr(> t)$ for <code>tmin_c</code>
What is R^2 ? Is the model practically useful?	Multiple R-squared
What are the units of b_1 ?	°C max per °C min
Are the assumptions plausible?	Diagnostics (Section 3)

This four-question checklist works for **any** SLR model.

Regression output: The checklist

Every time you fit a regression, answer four questions before reporting:

1. **Is the slope significant?** p-value for $b_1 < 0.05$? (F-test p-value < 0.05 ?)
2. **What is R^2 ?** What proportion of variance is explained? Is this practically useful?
3. **Can I trust the predictions?** Am I interpolating (within observed range) or extrapolating?
4. **Can I trust the model?** Are the LINE assumptions reasonably satisfied? (*Section 3*)

What the model cannot do

Even a well-fitted model has limitations:

- Assumes **linearity**: if the true relationship curves, the model is systematically wrong
- A **significant slope** does not prove causality: correlation $\neq$ causation (Section 1)
- **Outliers** can heavily influence b_1 and b_0
- A **high R^2** does not mean the model is correct. A model can fit well and still violate LINE assumptions
- Valid only in the **range of observed X**. Do not extrapolate without caution

This is the framing for Section 3. We are not looking for a perfect model.

“All models are wrong, but some are useful.”

George E. P. Box

Section 3 is not about finding a *perfect* model.

It is about checking whether the violations are *severe enough to undermine* our conclusions.

Can we trust the model?

The LINE assumptions

For the t-tests, CIs, F-test, and prediction intervals to be valid, four assumptions must hold:

L = Linear

The relationship between Y and X is linear (not curved).

N = Normal

The residuals are approximately normally distributed.

I = Independent

The residuals are independent of each other.

E = Equal variance

The variance of residuals is constant across all values of X (homoscedasticity).

Warning

If any assumption is **severely** violated, inferences (p-values, CIs, predictions) may be unreliable.

Mild departures, especially with large n , are usually tolerable.

Checking L: Linearity

Before fitting: look at the scatter plot.

- Does the cloud follow a straight line, or does it curve?
- If the scatter shows clear curvature → a linear model is inappropriate from the start.

After fitting: look at the residuals vs fitted plot (slide 48).

- A good plot shows a **random horizontal scatter** around zero, with no curve.
- A banana-shaped or U-shaped pattern in the residual plot means the linear model has missed systematic non-linearity.

For our examples:

- Sales ~ Advert: Scatter on slide 10 looks broadly linear ✓
- tmax_c ~ tmin_c: Scatter on slide 5 looks linear ✓

Checking I: Independence

Independence depends on **how the data were collected**. It's a design question, not a plot question.

Sales data:

36 independent advertising campaigns =
Each is a separate event

Reasonable to assume independence. ✓

Irish weather:

Daily records from the same station, 2000–2023.

Temperature today is correlated with temperature tomorrow → **autocorrelation**.

⚠ Independence assumption is likely violated.

For now: note it and continue. Formal time-series diagnostics (e.g. Durbin-Watson test) are beyond this module.

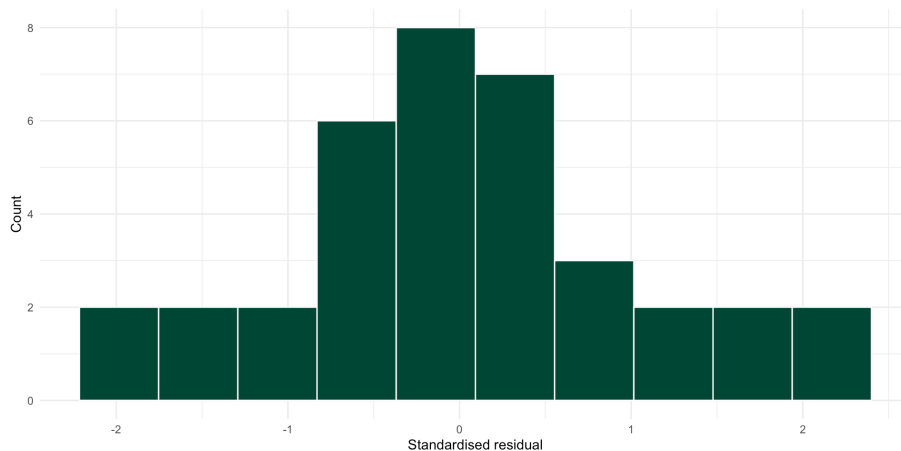
Note

Think about how the data were collected.

Time-series data and clustered data are common sources of dependence.

Checking N: Normality of residuals

```
1 z_res <- rstandard(reg1)
2
3 ggplot(data.frame(z_res), aes(x = z_res)) +
4   geom_histogram(fill = "#034638",
5                 colour = "white", bins = 10) +
6   labs(x = "Standardised residual",
7        y = "Count") +
8   theme_minimal()
```



Standardised residuals:

$$z_i = \frac{e_i}{\hat{\sigma}}$$

Divide each residual by the residual standard error $\hat{\sigma}$. In R, `rstandard()` applies a small additional leverage correction.

Should look approximately **bell-shaped, centred on zero**.

Watch for:

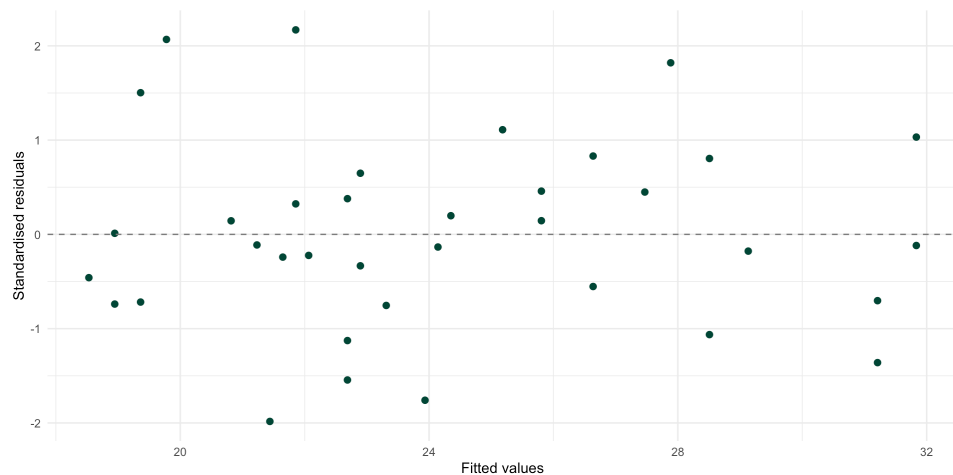
- Heavy tails → non-normal errors
- Strong skew → non-normal errors
- Bimodality → possible subgroups in the data

With large n : mild non-normality is not serious (CLT applies to estimators).

Checking E: Equal variance (homoscedasticity)

Plot the standardised residuals against the fitted values:

```
1 diag_df <- data.frame(fitted = fitted(reg1), resid = rstan  
2  
3 ggplot(diag_df, aes(x = fitted, y = resid)) +  
4   geom_point(colour = "#034638", size = 2) +  
5   geom_hline(yintercept = 0,  
6             linetype = "dashed",  
7             colour = "grey50") +  
8   labs(x = "Fitted values",  
9        y = "Standardised residuals") +  
10  theme_minimal()
```



What to look for:

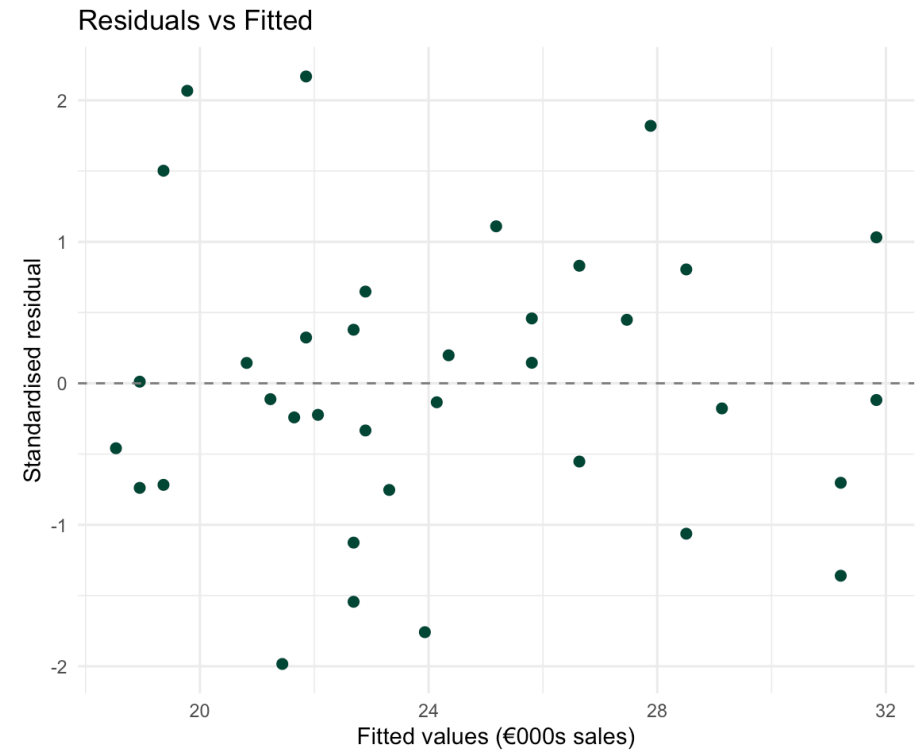
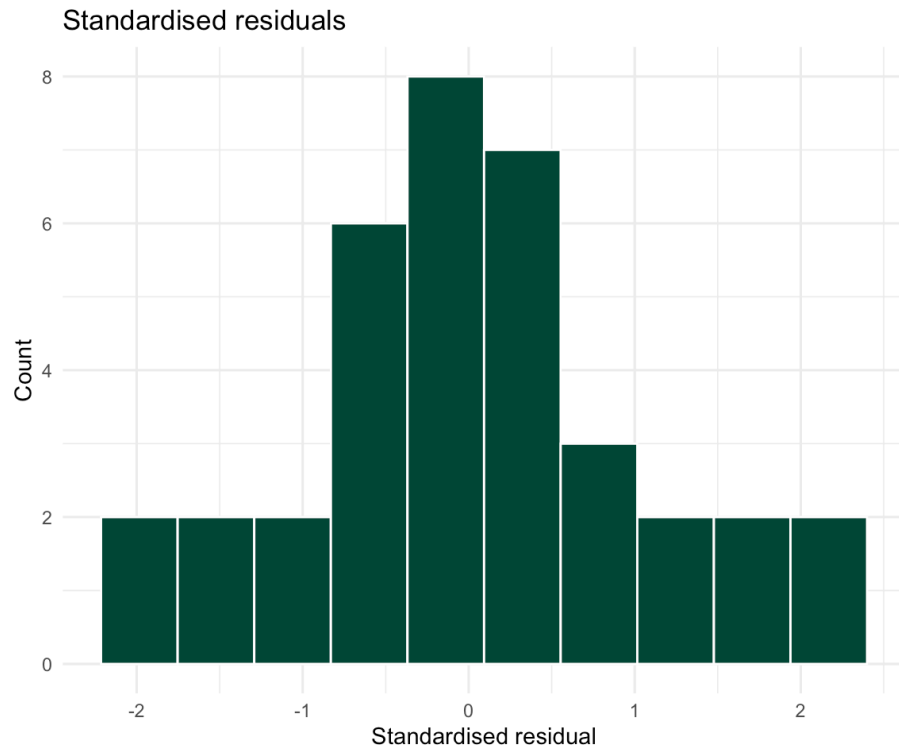
✓ **Good:** random scatter around zero = homoscedastic.

⚠ **Fan shape:** variance increases with fitted value → **heteroscedasticity** (E violated).

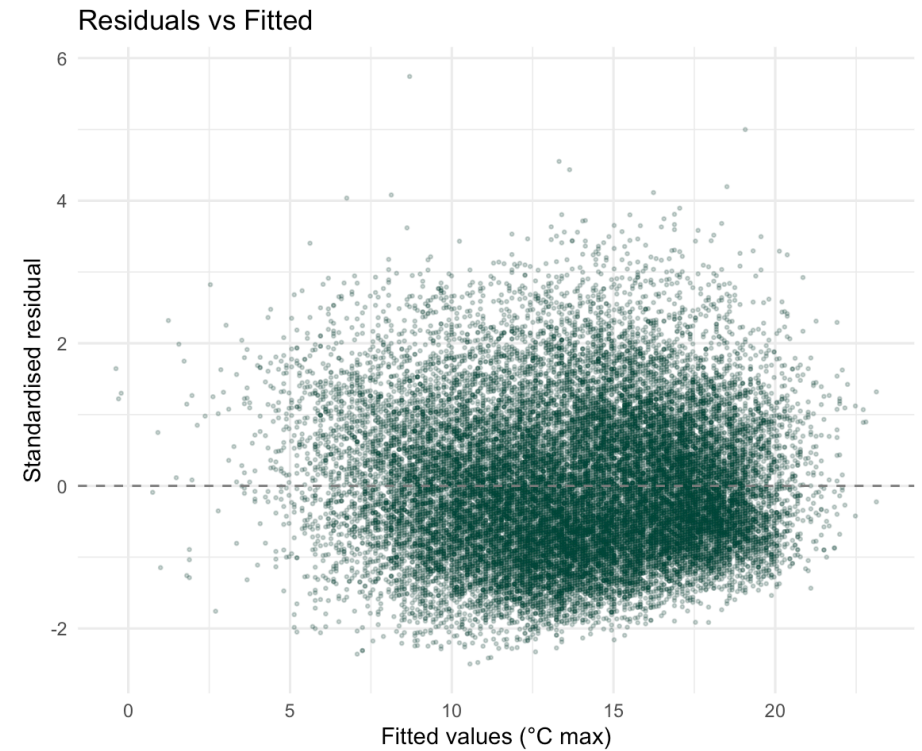
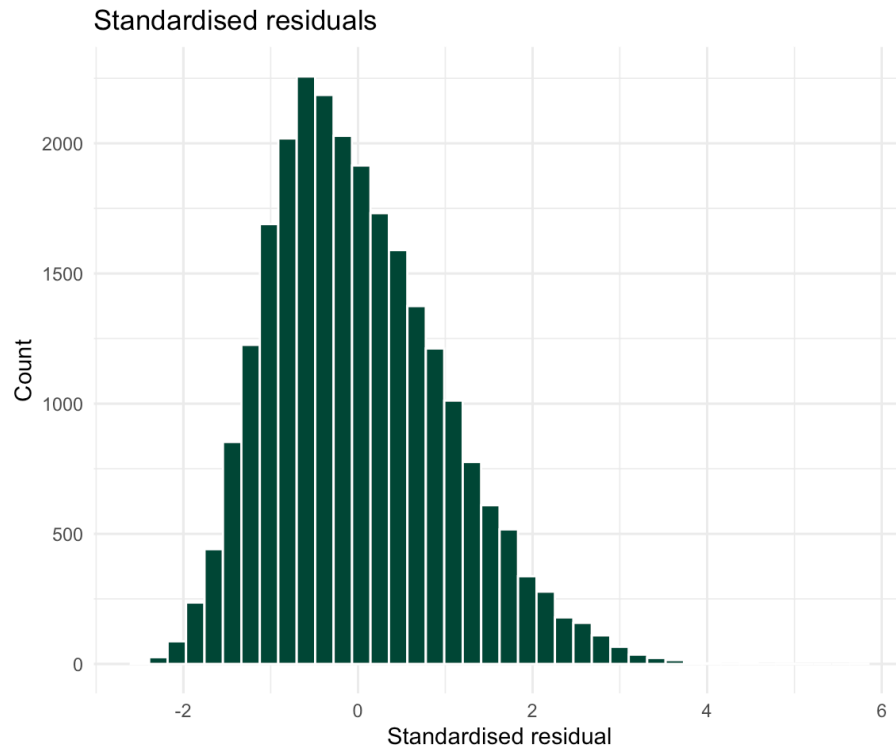
⚠ **Curved pattern:** systematic curve → non-linearity (L violated).

Heteroscedasticity means $SE(\hat{b}_1)$ is incorrectly estimated → p-values and CIs are unreliable even if the coefficient estimates are still unbiased.

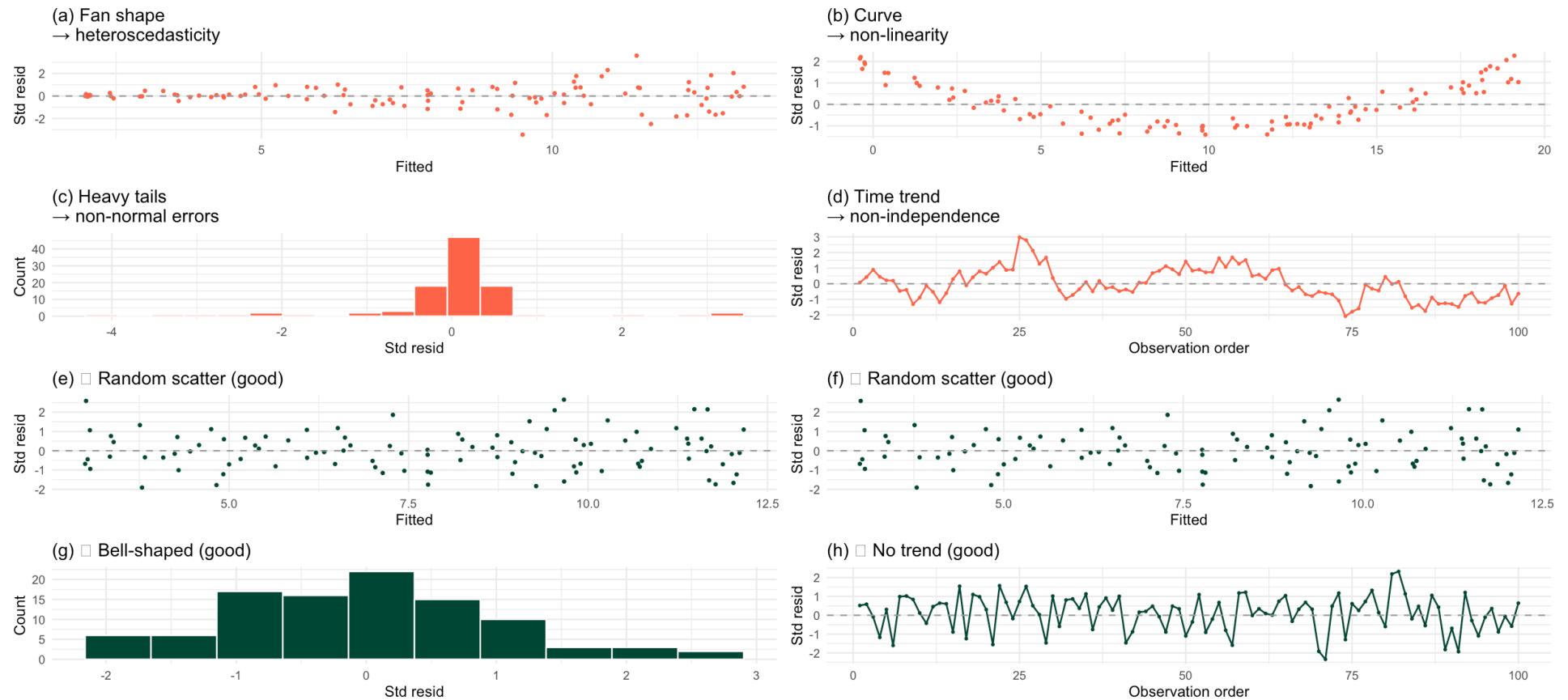
Running diagnostics in R: Sales data



Irish weather diagnostics



What violations look like



R functions

Function	Package	Purpose
<code>cor(x, y)</code>	base	Pearson correlation coefficient
<code>cor.test(x, y)</code>	base	Correlation with hypothesis test
<code>corrplot(M)</code>	corrplot	Visual correlation matrix
<code>pairs(df)</code>	base	Scatter plot matrix
<code>lm(y ~ x, data = df)</code>	base	Fit a simple linear regression model
<code>summary(model)</code>	base	Coefficients, R^2 , F-test
<code>anova(model)</code>	base	ANOVA table (SST, SSR, SSE)
<code>confint(model)</code>	base	Confidence intervals for coefficients
<code>predict(model, newdata)</code>	base	Fitted values or predictions
<code>plot(model)</code>	base	Four LINE diagnostic plots
<code>rstandard(model)</code>	base	Standardised residuals

Summary

Three things you can now do:

- 1. Measure association:** Pearson r (linear, continuous) or Spearman (ordinal, outliers); use `cor()`, `corrplot()`, `pairs()`; always plot first; correlation $\neq$ causation.
- 2. Fit and interpret a simple linear regression:** `lm()`, `summary()`, `anova()`, `confint()`, `predict()`; interpret the slope in units; report R^2 ; don't extrapolate.
- 3. Diagnose the model:** LINE assumptions; residuals vs fitted for L and E; histogram of residuals for N; think about data collection for I; use `rstandard()` and `fitted()`.